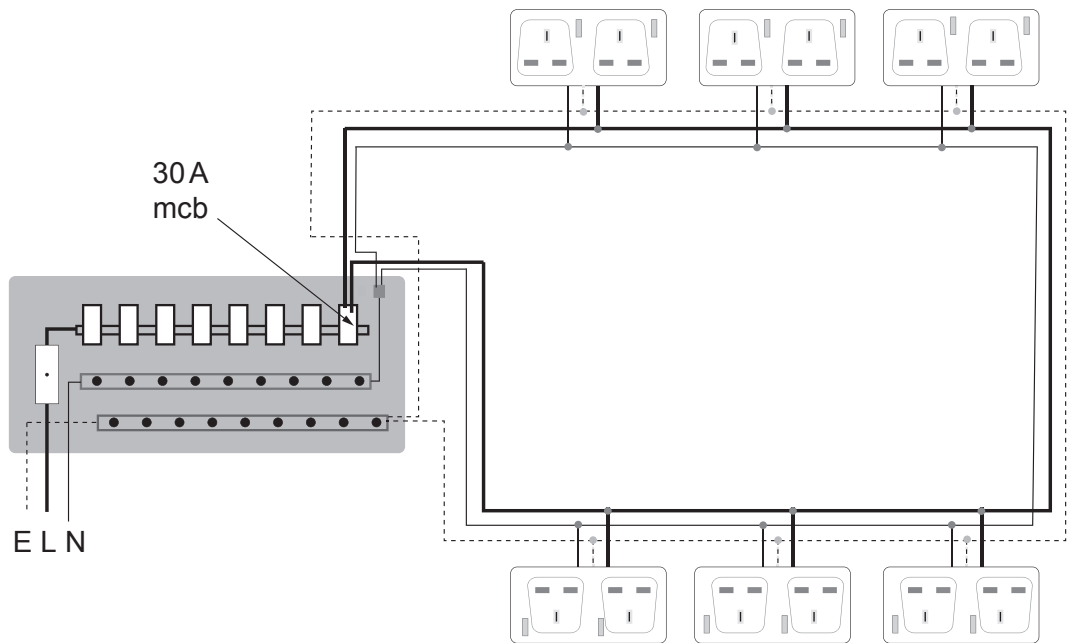


2. The diagram shows a common domestic ring main circuit.



(a) Give **two** advantages of having a ring main circuit in a house. [2]

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Examiner
only

(b) (i) State the function of the live wire. [1]

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.....

(ii) Describe the function of the earth wire. [2]

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(c) The following information is printed on the bottom of a domestic kettle:

TYPE: KBZ3001.BQ
230 V ~ a.c.
2760 W
CAPACITY 1.5L
SERIAL NUMBER 53023

(i) State the differences between alternating current (a.c.) and direct current (d.c.). [2]

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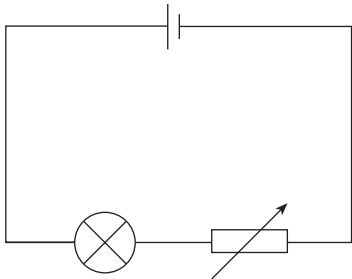
(ii) The voltage of the mains electricity supply in the U.K. is 230V. Use equations from page 2 and information on the kettle to calculate the resistance of the kettle heater. [4]

Resistance = Ω

11



3. (a) Students are investigating the current-voltage characteristic of a filament lamp. An incomplete diagram of their circuit is shown below.



- (i) **Add to the diagram** a correctly connected ammeter and voltmeter. [2]
- (ii) Describe how the circuit is used to obtain a series of measurements of current and voltage. [3]

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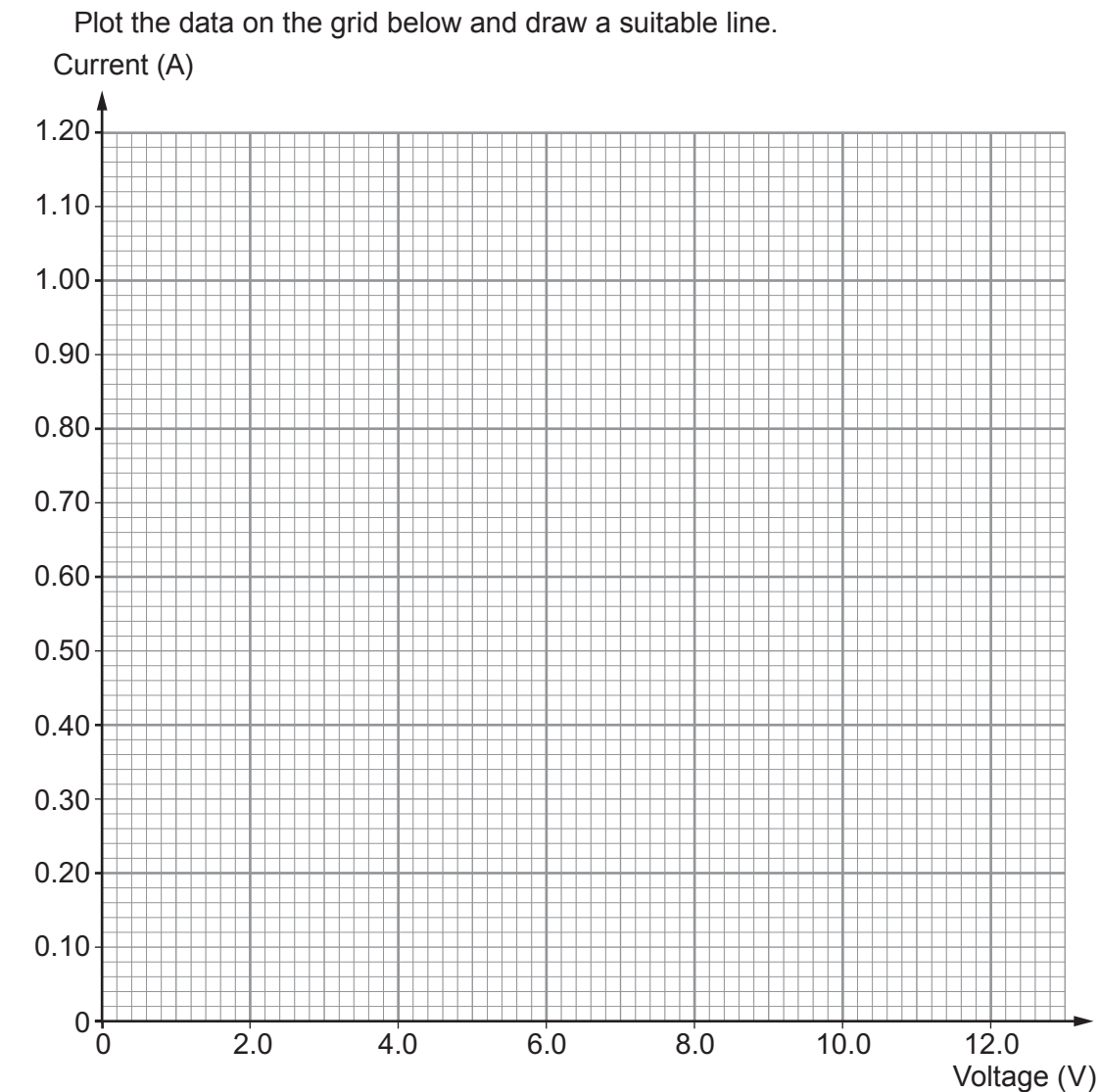
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- (b) The following data were collected using the circuit.

Voltage (V)	Current (A)
0.0	0.00
2.0	0.46
4.0	0.76
6.0	0.94
8.0	1.02
10.0	1.06
12.0	1.08



Examiner
only



(c) (i) Use the graph and the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp when the voltage across it is 3 V. [2]

Resistance = Ω

(ii) The students conclude that the **resistance** of the lamp increases as voltage increases. Explain how the data support this conclusion. [2]

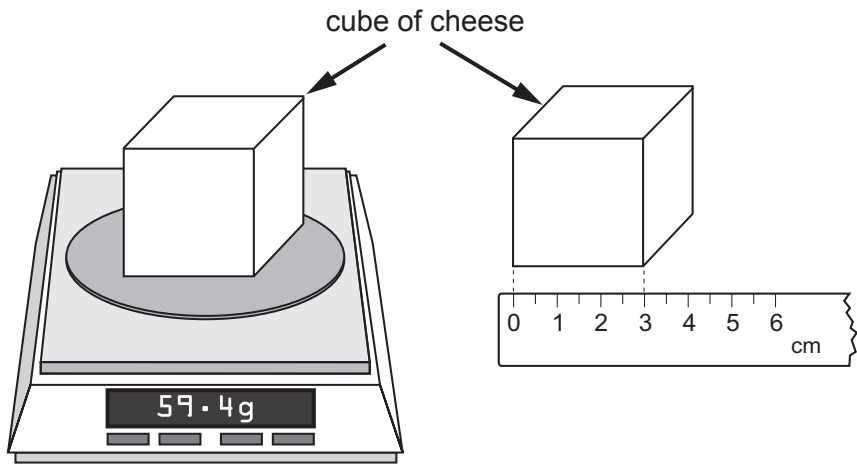
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4. A North Wales cheese producer checks the quality of their cheese by measuring its density. This can be carried out experimentally. The two diagrams show measurements that were carried out on a cube of cheese. The balance read 0.0 g before the cube of cheese was placed on it.



- (a) (i) Write down the mass of the cheese cube. g [1]
- (ii) Write down the length of one side of the cheese cube. cm [1]
- (iii) Use the equation:

volume of a cuboid = length × width × height

to calculate the volume of the cheese cube in cm³. [1]

Volume = cm³

- (iv) Use the equation:

density = $\frac{\text{mass}}{\text{volume}}$

to calculate the density of the cheese cube and state the unit. [3]

Density = Unit =



- (b) (i) A worker at the cheese factory suggests that the mass of the cheese cube should be measured more than once. State **two** reasons why this is good scientific practice. [2]

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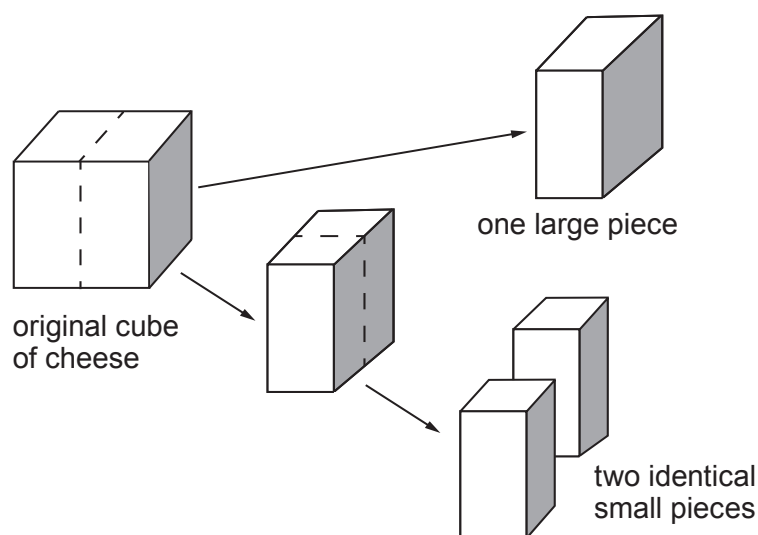
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- (ii) State how the **measurements** could be improved. [1]

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- (c) The cube of cheese used in the experiment is now cut into two identical halves. One of these pieces is then cut in half. The sample of cheese has been divided into three pieces, one large piece and two identical small pieces.



Tick (✓) the **two** correct statements.

[2]

One of the small pieces of cheese is $\frac{1}{3}$ of the mass of the large piece.

☐

One of the small pieces of cheese is $\frac{1}{3}$ of the volume of the large piece.

☐

The three pieces of cheese have the same density.

☐

The mass of the large piece of cheese is half the mass of the original cube.

☐

The mass to volume ratio for each of the three samples is different.

☐


5. The diagram shows an electric hand drier connected to a 230 V a.c. mains supply.



(a) Complete the following sentence by underlining the correct phrase. [1]

An a.c. or alternating current (**continuously changes direction / continuously changes frequency / continuously changes speed**).

(b) The electric hand drier has a label attached to it.

Electric hand drier	
Power = 2600 W	230 V a.c

(i) Each time the hand drier is used it blows out hot air for 15 seconds. Use an equation from page 2 to calculate the energy transferred by the hand drier if it is used **6 times**. [3]

Energy transferred = J

(ii) Use the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current when the hand drier is used. [2]

Current = A



Examiner
only

- (iii) The hand drier is connected to the mains with a standard three-core cable. Complete the table below with the value of the current in each wire in the connecting cable when the hand drier is operating normally. [2]

Wire	Current (A)
live	
neutral	
earth	

- (c) A fault develops and the live wire touches the metal case of the hand drier. Explain how a residual current circuit breaker (rccb) protects the user. [2]

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- (d) A salesperson for a replacement hand drier claims it will use fewer units of electricity each time it is used because its power is 1 300 W. When it operates it blows hot air for double the time. Explain if you agree with the claim of the salesperson. [2]

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12



6. Water waves on the sea where the water is shallower than $\frac{1}{20}$ of their wavelength are known as shallow water waves. The speed of shallow water waves is described by the equation:

$$v = 3.13\sqrt{d}$$

where v is the wave speed (in m/s) and d is the depth of the water (in m).
This equation applies to sea waves whose wavelengths range between 10 m and 150 m.

In regions of the sea where the depth is small, for example near the shore, the speed noticeably changes but the frequency of the waves remains constant.

A shallow water wave is an example of a transverse wave.

- (a) Describe what is meant by a transverse wave. [2]

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- (b) (i) Use the equation above to **complete the table below**. [2]
Space for workings.

Depth of water, d (m)	$\sqrt{d}$	Wave speed, v (m/s)
0	0	0
0.5	0.71	2.21
1.0	1.00	3.13
1.5		3.83
2.5	1.58	4.95
3.0	1.73	5.42
3.5	1.87	
4.0	2.00	6.26



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(ii) Chris suggests that if the depth of the water increases four times, the wave speed doubles. **Use data in the table opposite** to explain whether or not this statement is true. [2]

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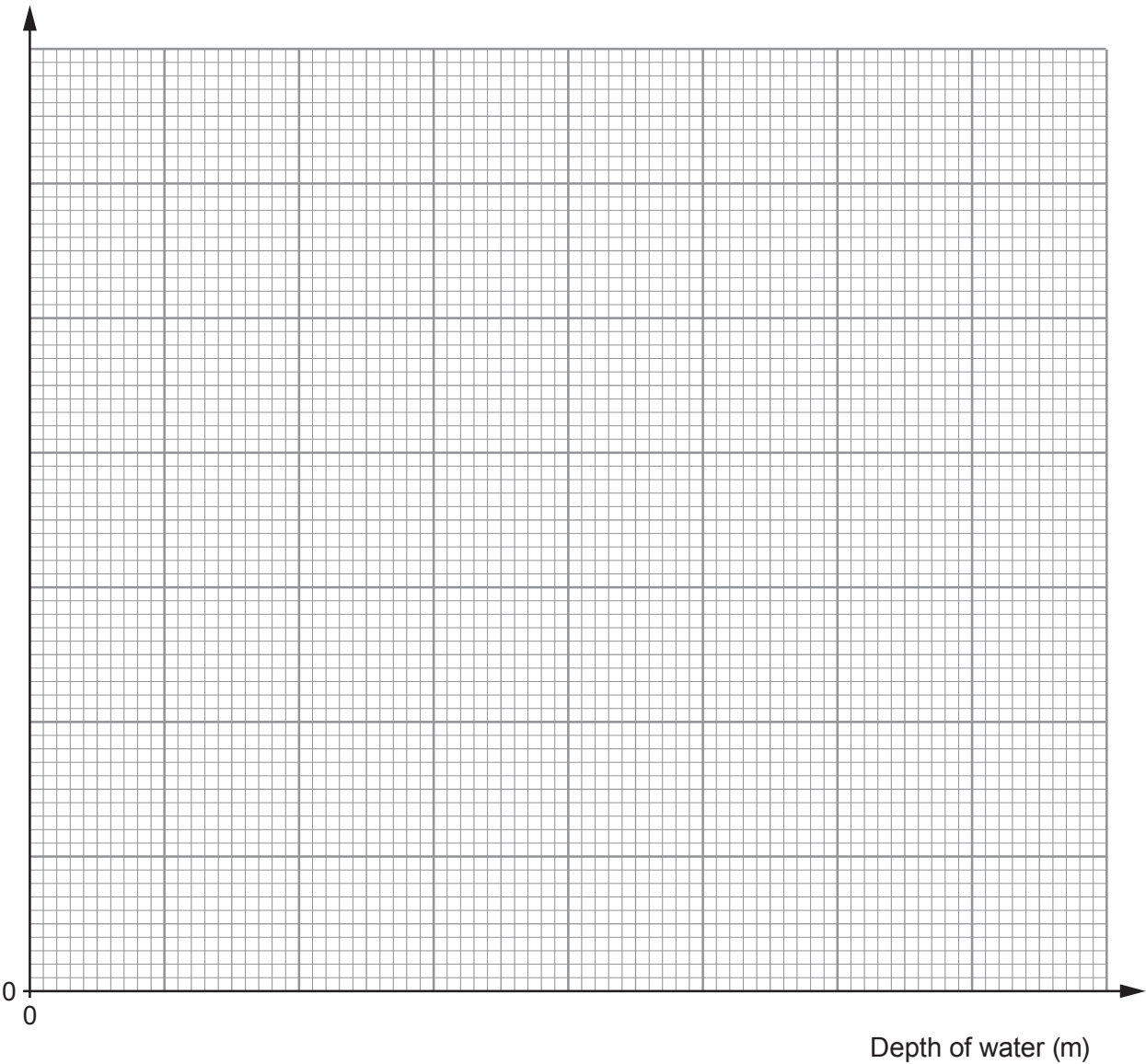
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(iii) Plot the data on the grid below and draw a suitable line. [4]

Wave speed (m/s)



Examiner
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(c) (i) Use the graph and the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of water waves that have a frequency of 0.2 Hz in water that is 2.0 m deep. [3]

Wavelength = m

(ii) Chris now suggests that as the depth increases, the wavelength decreases. Explain whether this statement is correct. [2]

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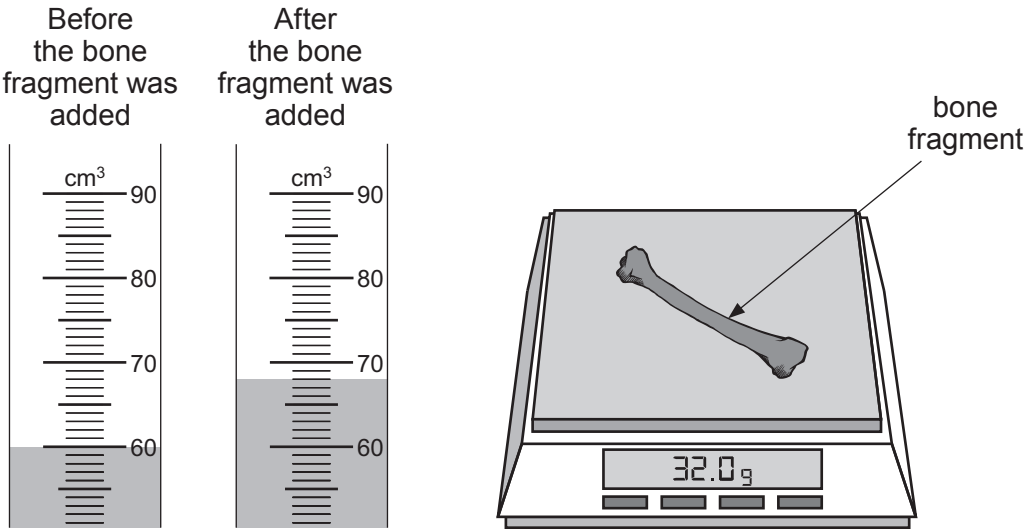
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END OF PAPER



Answer all questions.

1. The diagrams below show the apparatus used to find the density of a fragment of bone. The diagram of the measuring cylinder shows the water level before and after the bone fragment was added.



- (a) (i) Use the information in the diagrams to **complete the table below**. [2]

Mass of bone	 g
Volume of bone	 cm ³

- (ii) Use an equation from page 2 to calculate the density of the bone fragment. [2]

Density = g/cm³

- (b) Explain, in terms of particles, why bone has a greater density than water. [2]

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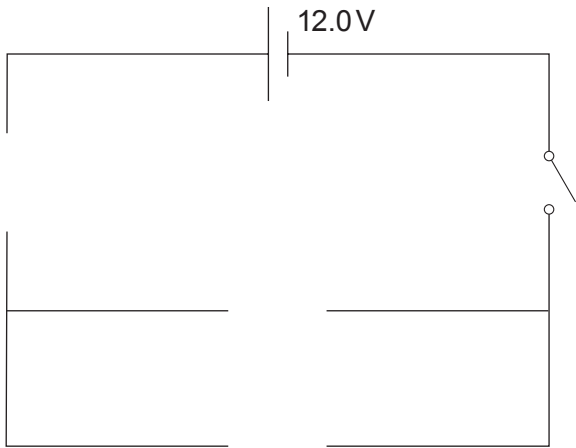
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6. A thermistor is a type of resistor whose resistance changes with temperature. Thermistors are used as temperature sensors.

To investigate a thermistor, students set up the following circuit to take measurements of current through the thermistor and the voltage across it.



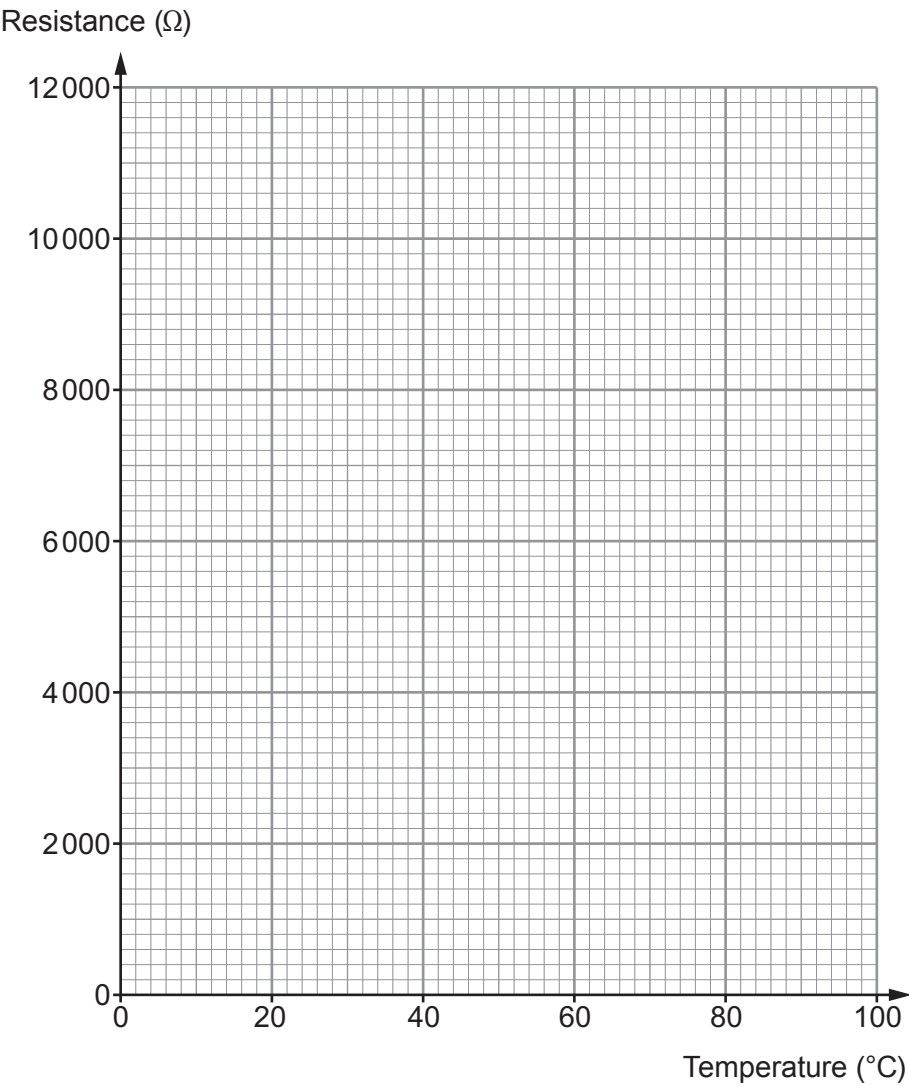
- (a) **Complete** the circuit diagram using the correct circuit symbols. [3]



(b) The students place the thermistor in water. They take current and voltage readings at different temperatures. These are used to calculate the resistance of the thermistor at each temperature. The results are shown in the table below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (Ω)
20	1.0	12.0	12 000
40	2.2	12.0	5 400
60	4.8	12.0	2 500
80	8.6	12.0	1 400
100	20.0	12.0	600

(i) Plot the data on the grid below and draw a suitable line. [3]



Examiner
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(ii) Describe what happens to the resistance of the thermistor as the temperature increases. [2]

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(c) (i) Use your graph to find the resistance of the thermistor at 50 °C. [1]

Resistance at 50 °C = Ω

(ii) Use an equation from page 2 to calculate the current through the thermistor at 50 °C. [3]

Current = A

(d) A temperature sensor needs to vary in resistance by at least 3600 Ω as the temperature changes from 40 °C to 80 °C. Use the results from the experiment to explain whether the thermistor used by the students would be suitable. [2]

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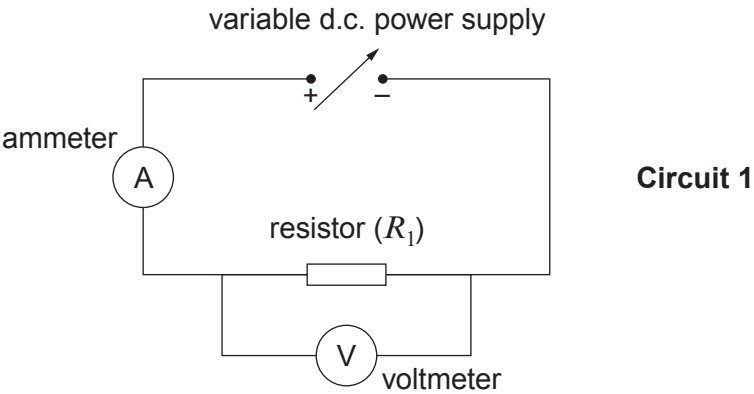
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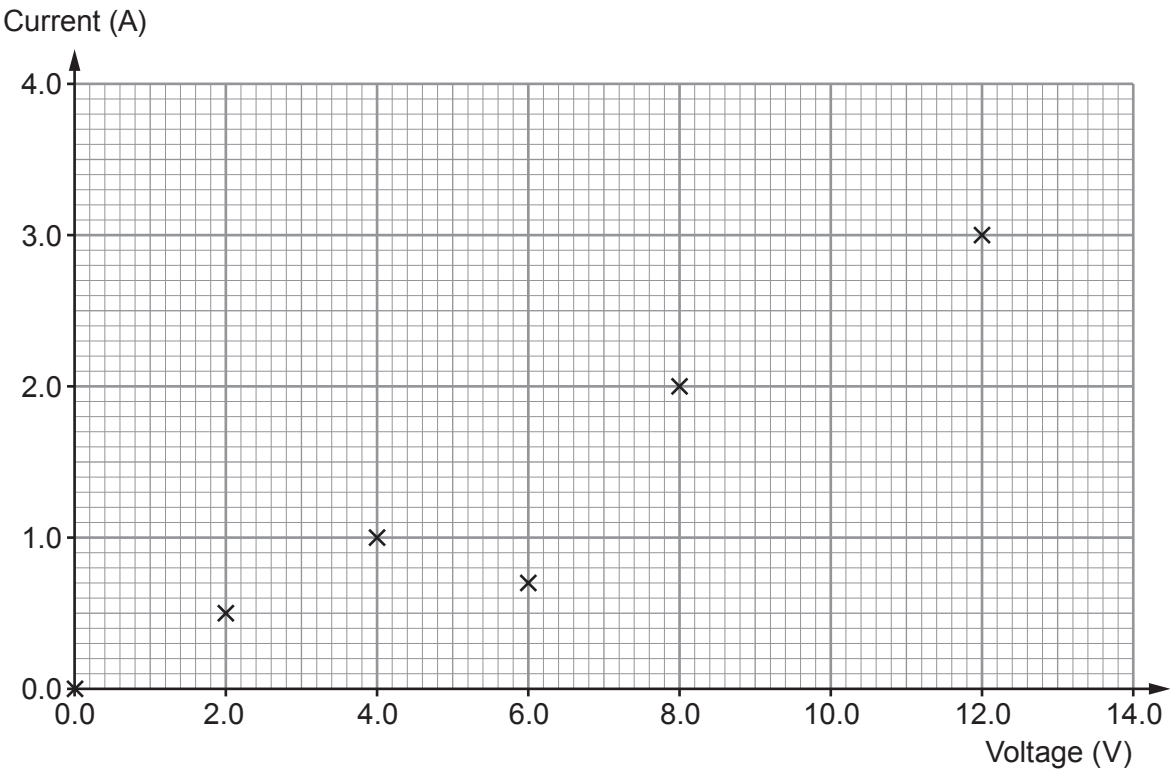


Answer **all** questions.

1. Paul investigated how the current changes with voltage for a resistor (R_1). The circuit he used is shown (**Circuit 1**).



- (a) Name the device he used to measure the current. [1]
- (b) Paul's results are shown on the graph.



- (i) On the graph, **circle** the anomalous point. [1]
- (ii) **Complete the graph** by drawing a suitable straight line. [1]



Examiner
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(iii) I. Use the graph to state the current at 12.0 V. [1]
Current = A

II. Use the equation:
$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

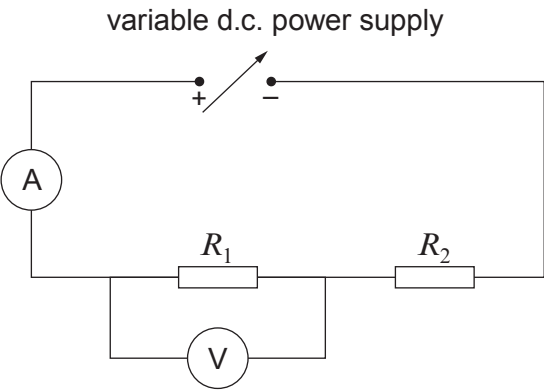
to calculate the resistance of the resistor (R_1) at 12.0 V. [2]

Resistance of R_1 = Ω

(iv) State what the graph shows about the resistance of the resistor (R_1) as the voltage changes. [1]

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(c) The original circuit is then adapted by adding a second, **identical** resistor (R_2) in series with R_1 .
This circuit is shown below (**Circuit 2**).



Complete the following sentences by underlining the correct word or phrase in each bracket. [3]

The total resistance in circuit 2 is (**double / the same as / half**) the total resistance of circuit 1.

When the d.c. power supply is set at the same voltage, the voltmeter reading in circuit 2 is (**double / the same as / half**) the voltmeter reading in circuit 1.

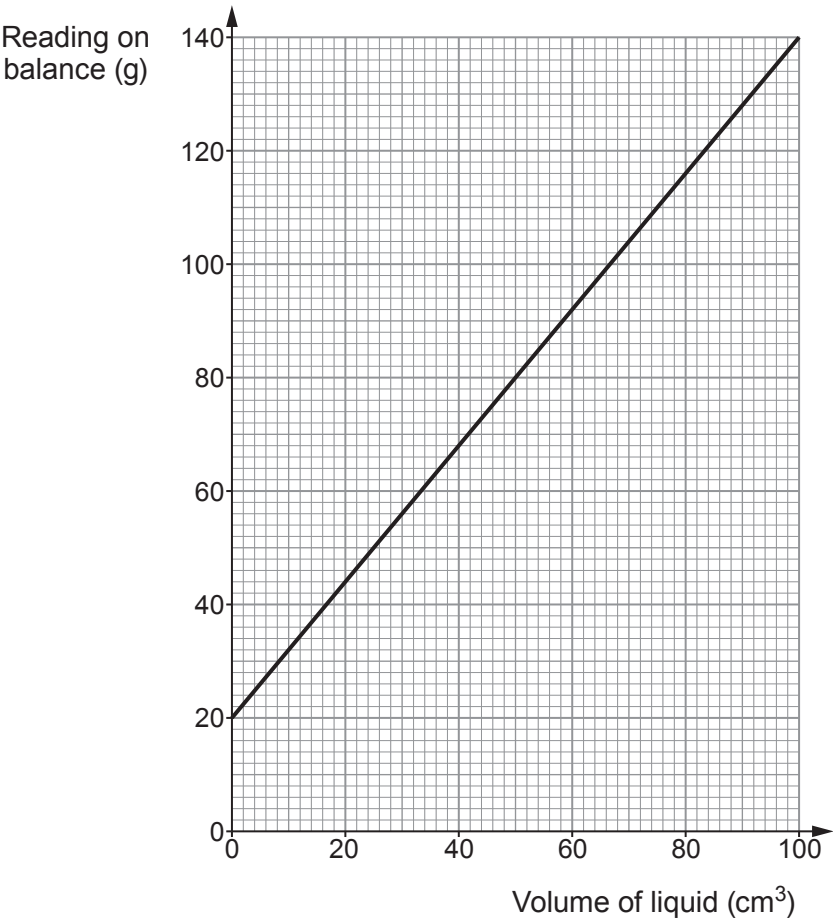
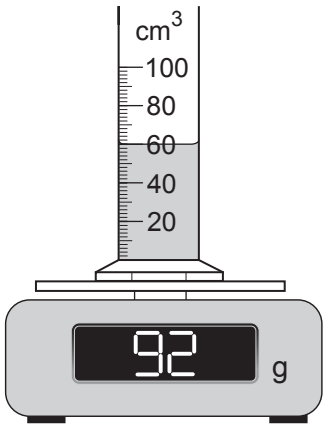
The current in circuit 2 is (**double / the same as / half**) the current in circuit 1.

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3. Kylie carries out an experiment, in class, to measure the density of a liquid. She places an empty measuring cylinder on a balance and pours the liquid into it. She records the reading from the balance for different volumes of liquid. Her results are shown in the table and on the graph.

Volume of liquid (cm ³)	Reading on balance (g)
20	44
40	68
60	92
80	116
100	140



Use the information above to answer the following questions.

- (a) State the mass of the empty measuring cylinder. g

[1]



Examiner
only

(b) Calculate the mass of liquid in the measuring cylinder when its volume is 100 cm³. [2]

Mass of liquid = g

(c) Use an equation from page 2 to calculate the density of the liquid that Kylie used and state the unit of your answer. [3]

Density =

Unit =

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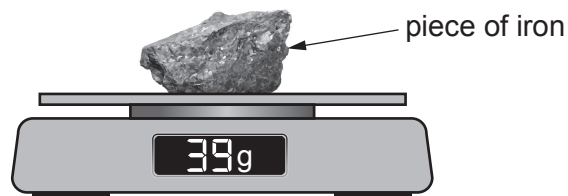
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3. Peter carries out an experiment to find the density of iron.

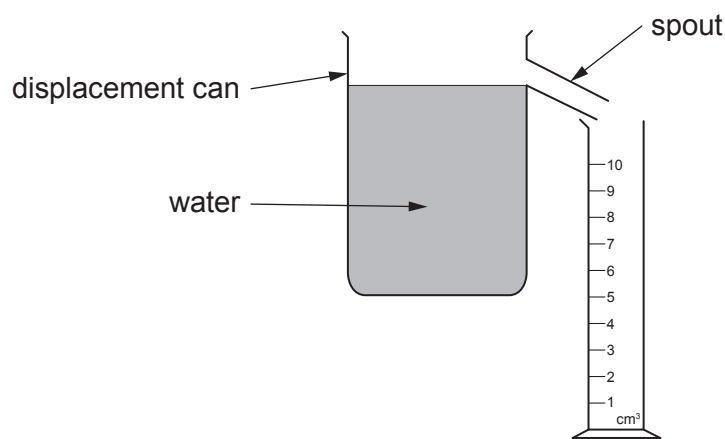
Stage 1

- He uses a balance to measure the mass of a piece of iron.



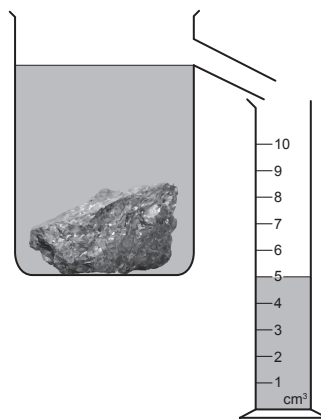
Stage 2

- He fills a displacement can with water.
- He waits until water stops flowing from the spout.
- He then places an empty measuring cylinder under the spout.



Stage 3

- He carefully lowers the iron into the water.
- He collects the water displaced in the measuring cylinder.



Examiner
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Use information from the diagrams to answer the following questions.

(i) State the mass of the iron. mass = g [1]

(ii) State the volume of the iron. volume = cm³ [1]

(iii) I. Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of iron. [2]

density =

II. Underline the correct unit that should be used for the density above. [1]

g/cm³ **g/cm²** **cm²/g** **cm³/g**

(iv) State **one** change to the apparatus that would improve the accuracy of the measurement of the **mass** of the iron. [1]

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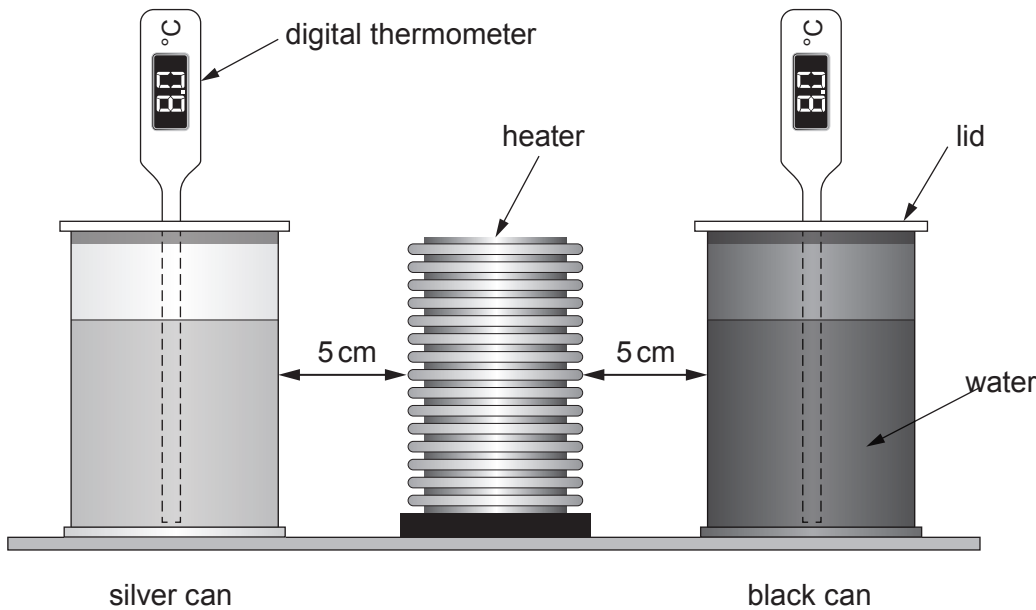


5. A pupil sets up an experiment to investigate the absorption of heat radiation.

The diagram shows the apparatus set up **before** the heater was turned on for 8 minutes.

Two identical aluminium cans were both filled with 200 cm³ of water and a digital thermometer was inserted through a small hole in the lid.

One can was painted black and the other painted silver.



(a) In the experiment several variables were controlled in order to make a valid conclusion.

Use the information above to describe how this was achieved.

In your answer include

- **three** variables that were controlled
- an explanation of why each was controlled.

[6 QER]

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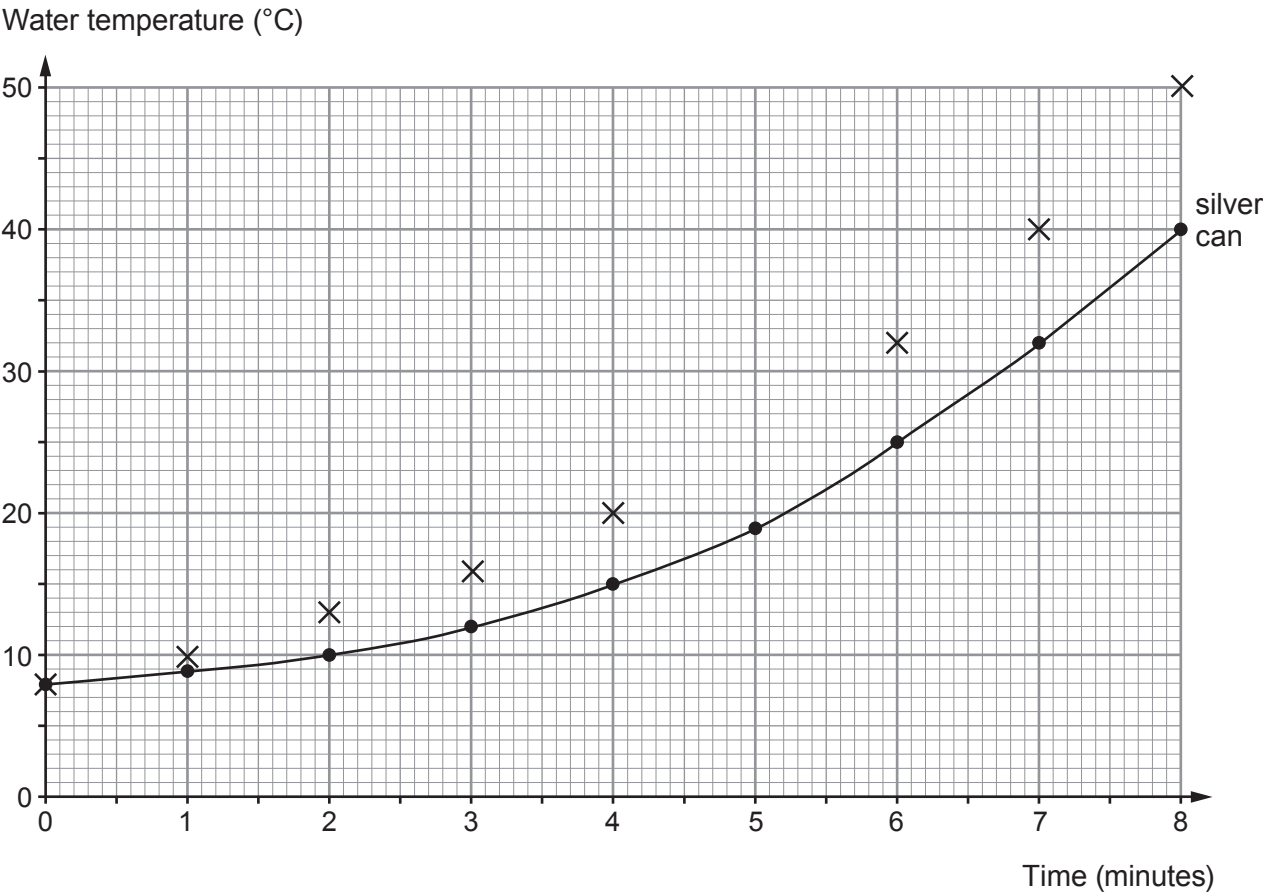
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(b) Each minute, the temperature of the water in each can was measured.

The data, for the silver can, have been plotted on the grid below and its best fit curve drawn.

Apart from one result, the data points for the black can were also plotted on the grid.



(i) At **5 minutes** the water temperature in the black can was **25°C** .

Plot this missing data point on the grid above **and** add the line of best fit. [2]

(ii) Compare the temperature rise in the black can with the temperature rise in the silver can.

Explain your answer in terms of heat radiation. [2]

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3. (a) A student is investigating the current-voltage characteristic of a filament lamp. In the space below, **draw a circuit diagram** of the circuit she would use. [3]

(b) The student collects the following results.

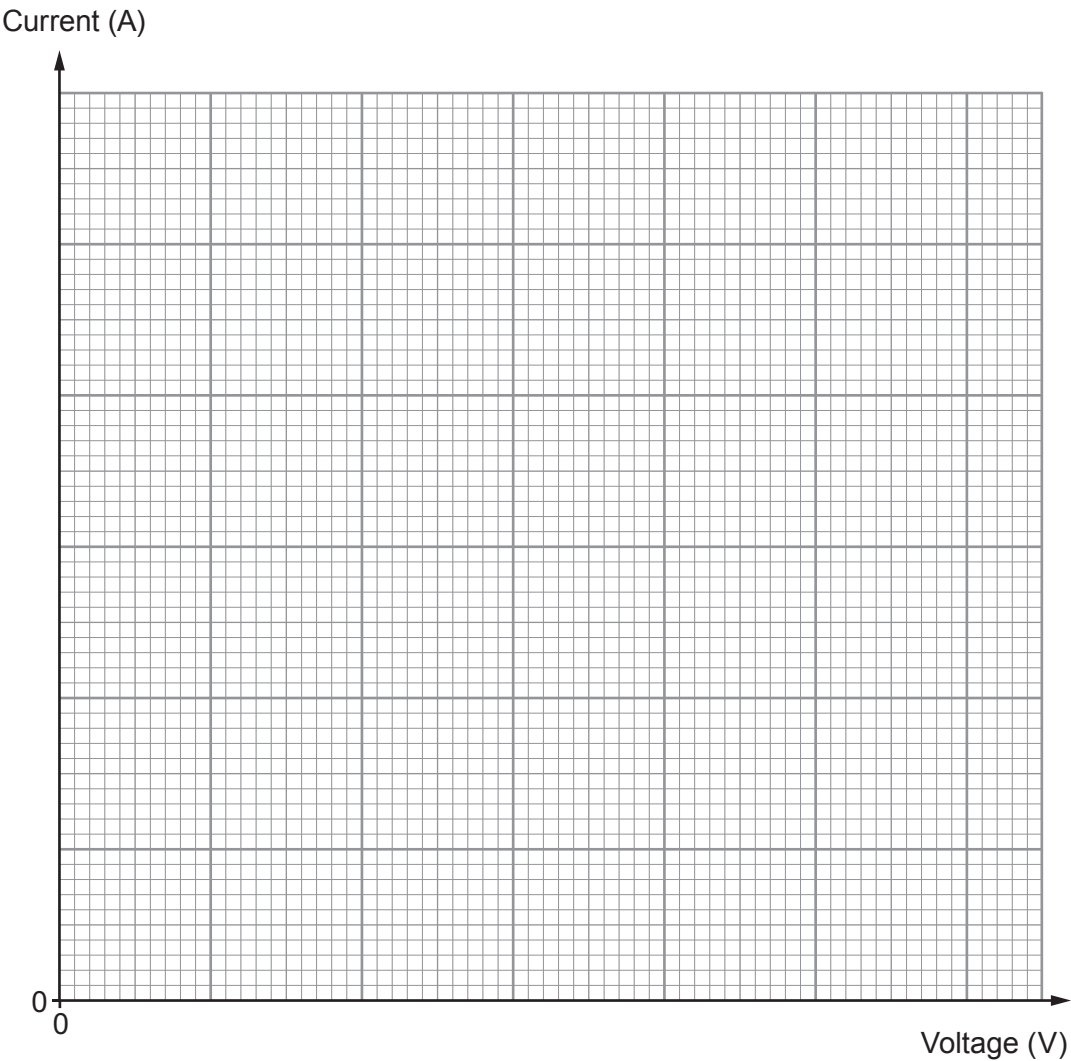
Voltage (V)	Current (A)
0.0	0.00
2.0	0.40
4.0	0.76
6.0	0.94
10.0	1.08
12.0	1.10



Examiner
only

(i) Plot the data on the grid below and draw a suitable line.

[3]



(ii) She concludes that the resistance of the lamp decreases as voltage increases. Explain using calculations, whether you agree or disagree with this conclusion. [3]

Space for calculations

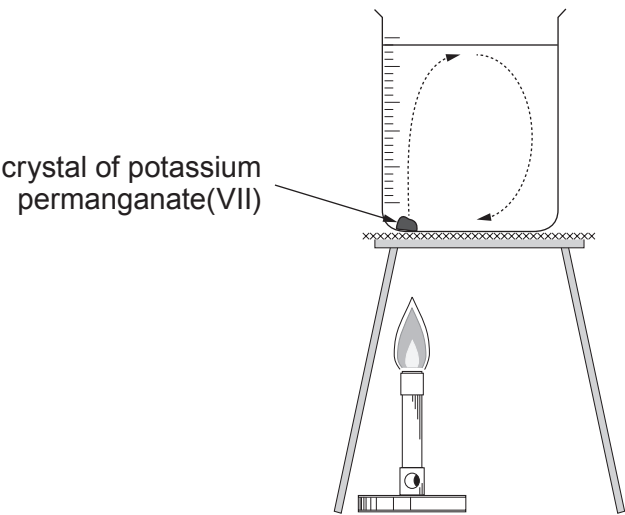
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5. A class carries out experiments to study heat transfer in solids and liquids.

- (a) In one experiment, the process of convection is shown by colouring some water with potassium permanganate(VII) crystals in the bottom corner of a beaker. The water is then heated and a clockwise convection current is set up.



Explain why the coloured water moves in the way described above.

[2]

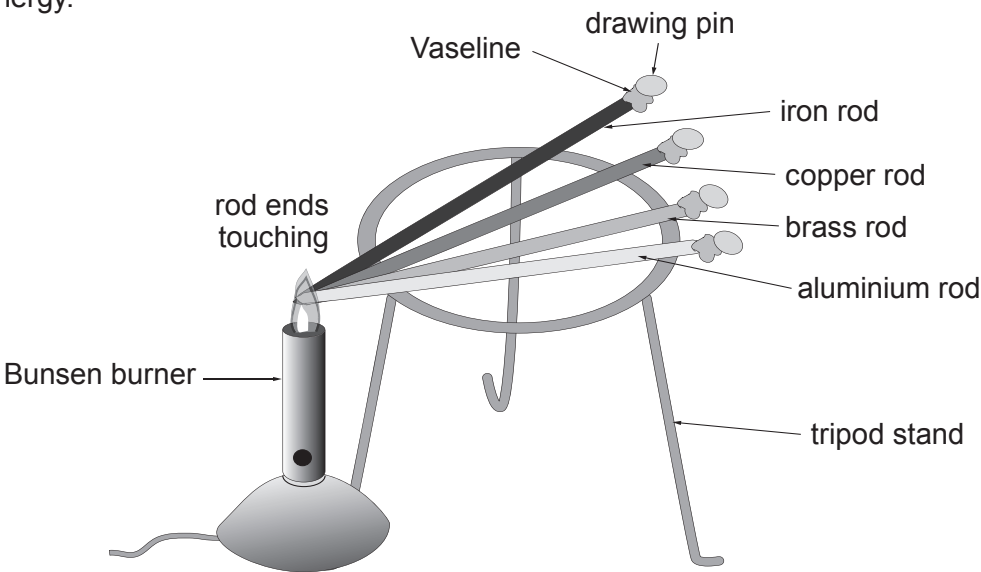
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- (b) In another experiment, four metal rods are tested to compare their ability to conduct heat energy.



Examiner
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The outcome of the experiment is as follows:

- copper (best conductor)
- aluminium
- brass
- iron (poorest conductor).

Explain in detail, the process of conduction in **copper** in terms of the motion of **particles**.
[6 QER]

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Examiner
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3. Describe how the density of a small pebble, which is not a regular shape, can be determined. [6 QER]

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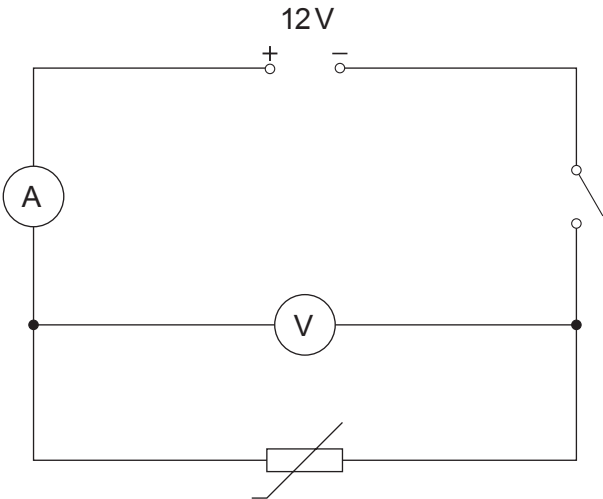
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4. A thermistor is a type of resistor whose resistance depends on temperature. Thermistors are often used as temperature sensors in applications where we want to monitor or control temperature. A group of students immerse a thermistor and a thermometer in water in order to determine the resistance at different temperatures. The circuit they use is shown below.



Their results are given below.

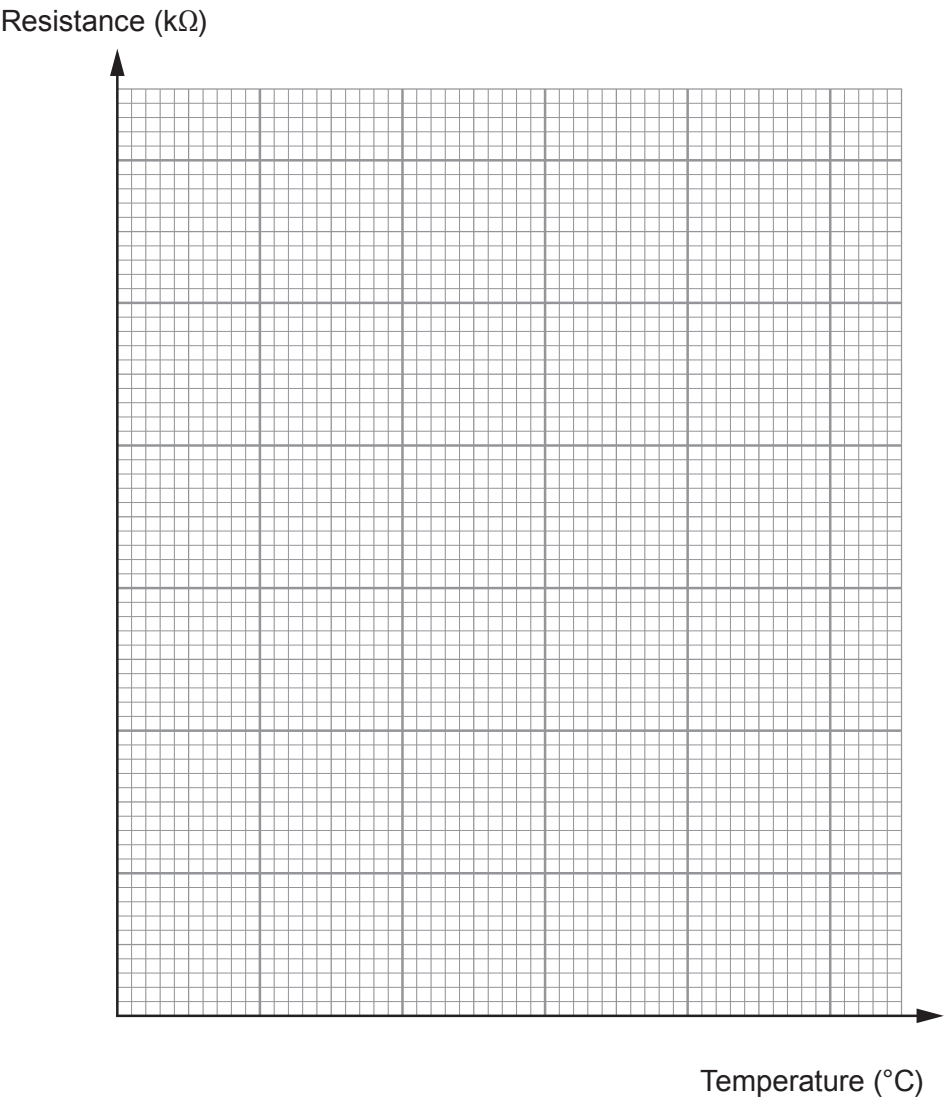
Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	0.97	12.0	12.4
40	2.22	12.0	5.41
60	4.80	12.0	2.50
80	9.23	12.0	1.30
100	17.14	12.0	0.70



Examiner
only

(a) (i) Plot the data on the grid below and draw a suitable line.

[4]



(ii) Describe the relationship between temperature and resistance.

[2]

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Examiner
only

- (b) (i) A resistor of resistance $5\text{ k}\Omega$ is now added to the original circuit, in parallel with the thermistor which is at a temperature of 30°C . The same 12 V power supply is used. Use your graph and equations from page 2 to calculate the total current in the circuit. [5]

Current = A

- (ii) Kim suggests that the current through both the resistor and the thermistor will decrease as the thermistor cools down. Explain whether or not she is correct. [3]

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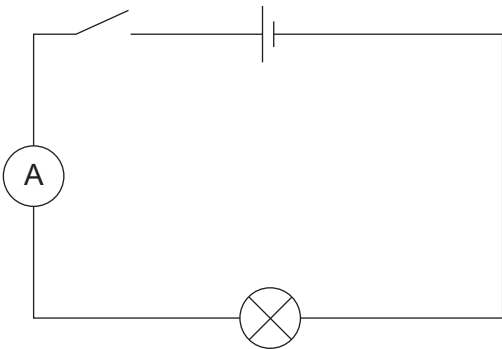
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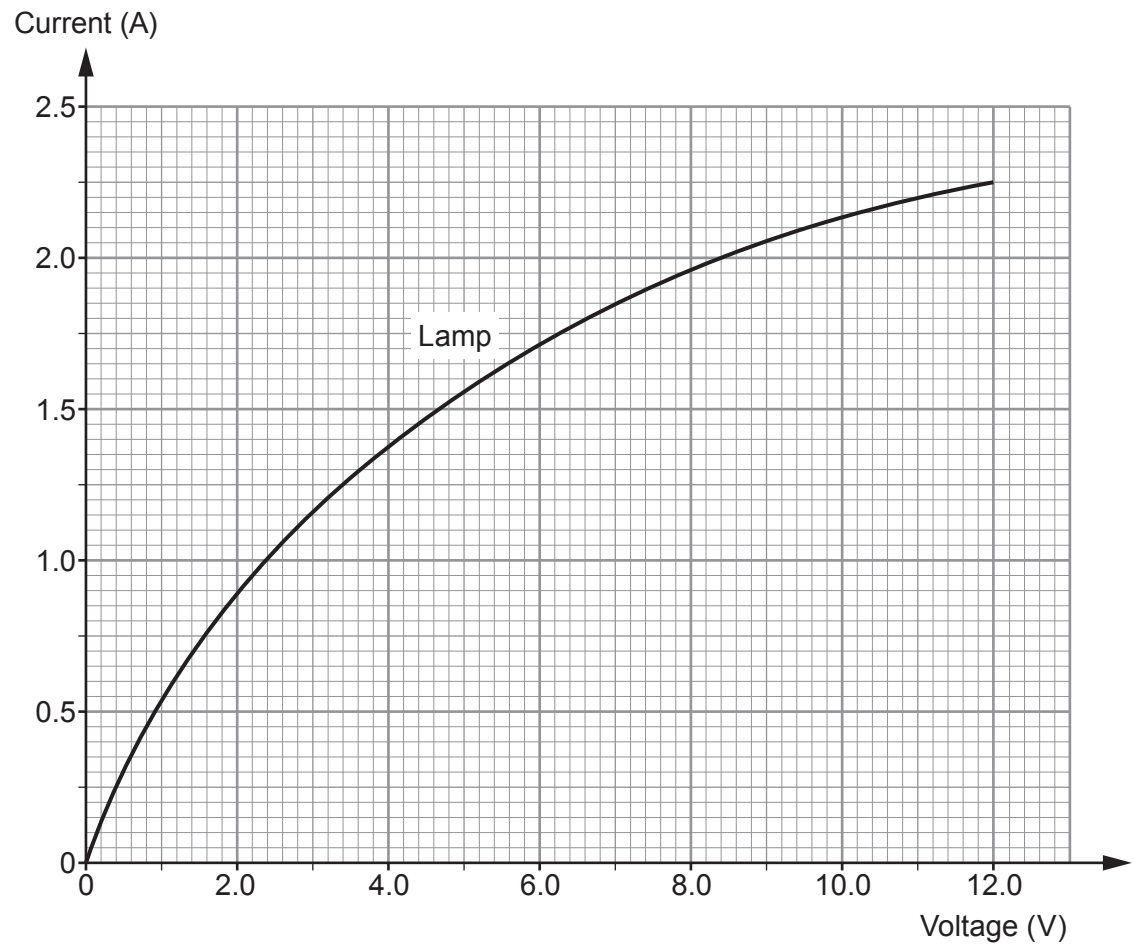


Answer **all** questions.

1. A group of pupils investigate the current-voltage (I - V) characteristics of different components.
- (a) The first component they investigate is a lamp.
Part of the circuit used is shown below.
Add a variable resistor **and** voltmeter to the circuit diagram. [2]



- (b) They draw a graph from their results for the **lamp**. It is shown below.



Examiner
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- (i) One student suggests that as the current through the lamp **doubles** the voltage **triples**.
Use pairs of data within the range 0.5 A to 2.0 A from the graph to explain whether you agree with the student. [3]

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- (ii) Use the equation:
- power = voltage × current
- and information from the graph to calculate the **maximum** power produced by the lamp. [3]

Power = W

- (c) The experiment is repeated with a 6 Ω resistor but the results are lost.

- (i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current through the 6 Ω resistor at 12 V. [2]

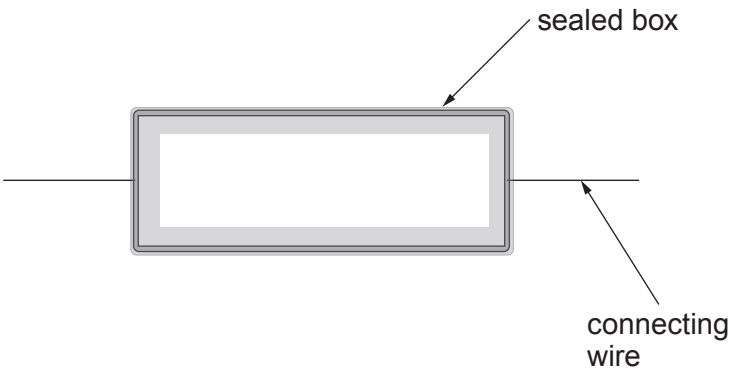
Current = A

- (ii) **Draw the line for this resistor** on the grid on the previous page. [1]

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(d) The students are given a sealed box containing another component.



They are asked to confirm whether the hidden component is a diode.

(i) Describe how they would use the circuit shown in part (a) to confirm whether or not it is a diode. [3]

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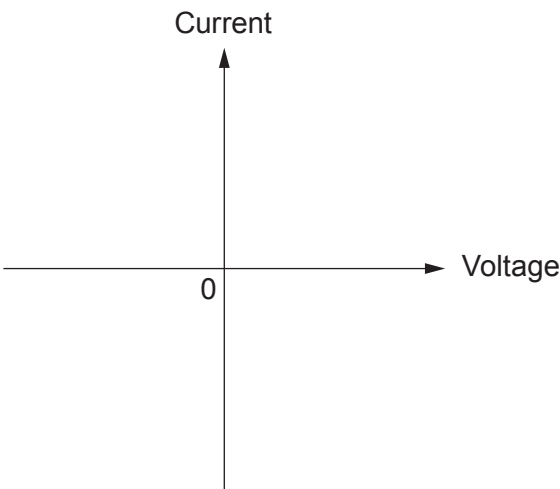
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(ii) **Sketch** the I - V graph that you would expect to obtain for a diode on the grid below. [1]

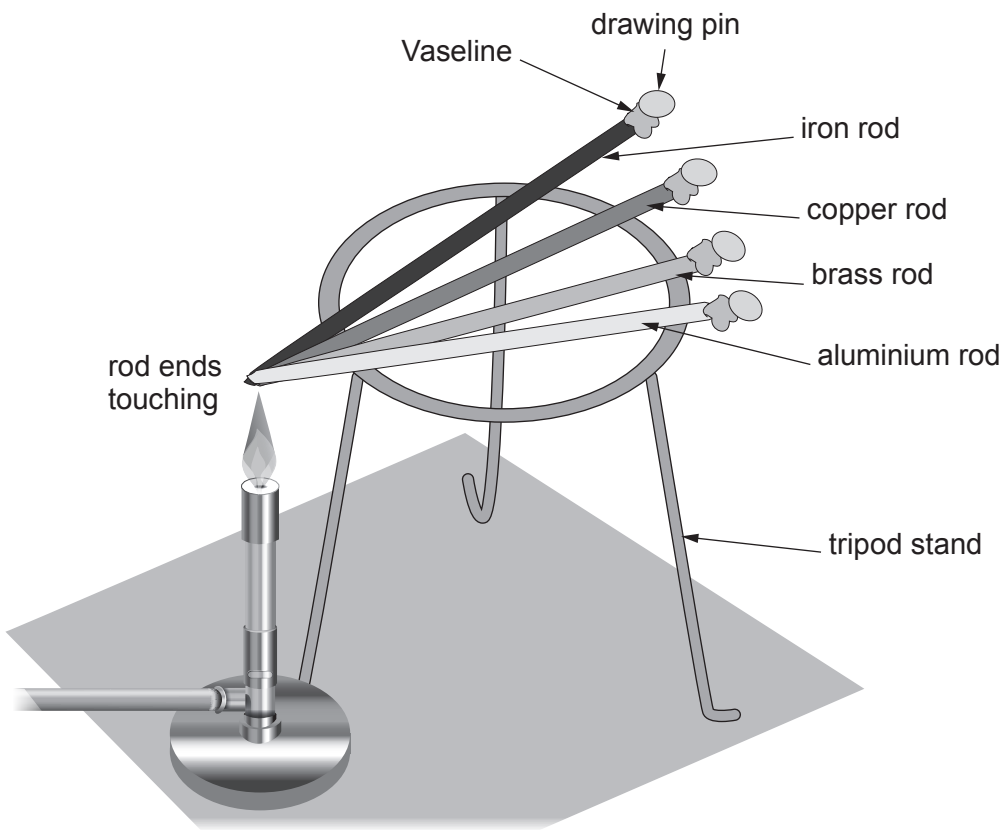


4. (a) (i) The apparatus shown can be used to compare how well different metals conduct heat.

Explain:

- how the apparatus is used
- how the results are used to show that copper is the best conductor.

[3]



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(ii) Describe, in terms of the behaviour of **particles**, how heat energy is conducted through the metal rods. [2]

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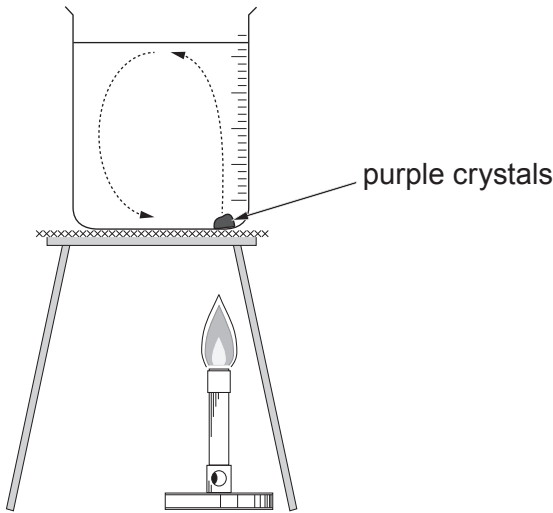
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(b) A teacher demonstrates the process of convection in liquids.

Some purple crystals are placed in the bottom corner of a beaker that is filled with water.

The water is then heated and a current of coloured water is seen to move as shown in the diagram below.



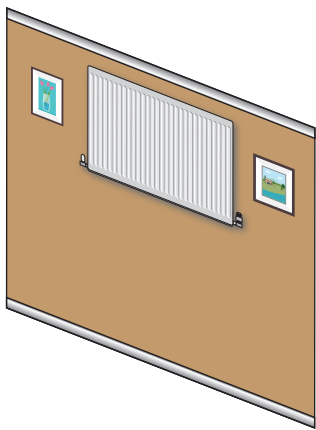
(i) Explain why the coloured water moves in the way shown above. [2]

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(ii) Explain why it is not suitable to put a hot radiator at the top of a wall to warm the room. [2]



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